

Certified analytical bounds for hypersonic reachable sets

A measure-theoretic profile decomposition in exact interval arithmetic

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Abstract

The reachable set of a nonlinear control system can be embedded in a space of occupation measures, where the dynamics become a linear constraint and the reachable set is read off the support of a terminal measure. This research asks what that embedding yields under two departures from the standard program: when the vector field is not a function but a set-valued map, enclosed in exact interval arithmetic; and when the measure family loses tightness in a singular limit. Rescaled by the fast timescale, mass concentrates on a low-dimensional slow manifold, and the limit admits a profile decomposition in the manner of Lions' concentration alternative—a finite-dimensional skeleton, one boundary-layer bubble per event, and a compact residual of order $\sqrt{\varepsilon}$. Almost-orthogonality of the scaling groups, transported from the Bahouri–Gérard construction to defect measures under bounded adversarial uncertainty, is what makes the decomposition a bound rather than an ansatz. The constants that size it are required in closed form, derived from the physical constraint set and the aerodynamic interval widths rather than from a numerical solver. Hypersonic entry supplies the instance: a six-degree-of-freedom vehicle whose thin-atmosphere limit furnishes the singular perturbation, whose equilibrium-glide condition furnishes the slow manifold, and whose skip events furnish the bubbles. The result is a certificate rather than a computation: explicit algebraic bounds, valid uniformly over the entire admissible family, in the minimal physical state space.

1 The analytical problem

The object to be bounded is the reachable set of an atmospheric entry vehicle—every state it can be driven to under admissible control—which is what safety and threat analyses need enclosed. The flight-mechanics specifics are deferred: [Section 3](#) states why entry is the right instance and what its state space looks like, and the coupled dynamics—frames, state vector, translational and rotational equations, the aerodynamic bridge from attitude to force, and the slow-manifold condition—are carried in a companion appendix omitted from this copy. What matters first is that the problem is an analytical one before it is an aerospace one, and that it fails for two independent reasons—the dimension of the state, and the provenance of the guarantee.

1.1 Reachable sets as occupation measures

Embed the hybrid flight dynamics into a Banach space of occupation measures. The nonlinear flight mechanics $\dot{x} = f_q(x, c, t)$ become *linear*: the Liouville equation is a linear constraint on a measure μ over $[0, T] \times \mathcal{X} \times \mathcal{U}$, and $\text{supp } \mu_T$ outer-approximates the reachable set. So far this is the occupation-measure program of Lasserre, Henrion and collaborators [\[1, 2\]](#). The embedding buys linearity at the cost of uniqueness, and the way it is recovered is standard: a trajectory family satisfying the Liouville constraint lies in the convex hull of the generalized curves, so measures correspond to convex mixtures of trajectories rather than to one of them [\[3, 4\]](#). The same relaxation is what admits the interval envelope, and that is the first thing that has to be built rather than borrowed. Because the true aerodynamics are transcendental, they are enclosed in strict intervals, and a vector field selected from an interval is a set-valued map; the resulting differential inclusion is the object the measure must describe, not the nominal field [\[5, 6\]](#). Occupation-measure theory for inclusions of exactly this kind already exists and must be cited as prior art: it defines an occupational invariant measure without assuming solution uniqueness, which is the same move made here [\[7\]](#). What it assumes is worth stating plainly, because it is where the entry problem is likely to fail: the vector fields must satisfy a linear growth condition, and the singular-perturbation results additionally require a uniform bound on the fast state along the whole ε -family [\[7, 8\]](#).

The current framework embeds the flight mechanics $\dot{x} = f_q(x, c, t)$ as a linear Liouville constraint on an occupation measure μ over $[0, T] \times \mathcal{X} \times \mathcal{U}$. By bounding true non-polynomial aerodynamics within strict intervals, the vector field becomes a differential inclusion $\dot{x} \in \mathcal{F}(x, c, t)$. We must formulate a robust Liouville equation capable of acting on set-valued maps, and rigorously prove that the support of the resulting occupation measure, $\text{supp } \mu_T$, securely outer-approximates all possible trajectories generated by any vector field selected from the interval inclusion.

1.2 Why the guarantee must be exact

The standard polynomial barrier with coefficients obtained from an interior-point solver has analytical *form* but inherits the solver’s accuracy [\[9\]](#), so it is not a certificate; a numerical solver’s failure modes are documented, and they include the loss of strict feasibility on which every rounding-based scheme depends [\[10, 11\]](#). Provenance is not a matter of care. Interval arithmetic is the setting in which a guarantee has meant, for forty years, an enclosure rather than an approximation, and the property that makes an enclosure valid at all—inclusion isotonicity—is exactly what a reimplementaion loses first; not every law of real arithmetic survives the move to intervals, which is the original form of the wrapping effect [\[12, 13\]](#). Applied interval analysis supplies the machinery the pipeline would

be built from, and it is deliberately conservative: certified set inversion, deterministic intersection geometries, and a customized exact-arithmetic integrator to ensure floating-point inaccuracies do not compromise the final bounds [14].

The proposal demands a guarantee verified in exact arithmetic rather than produced by a numerical solver. Floating-point inaccuracies inherent to standard simulation tools compromise the analytical certificate. Must implement a custom, exact-arithmetic interval integrator (likely leveraging C/C++ or specialized rational arithmetic libraries). This pipeline must process the hybrid dynamics, manage the boundary-layer bubble transitions, and compute the intersection geometries deterministically to verify the final bounds.

2 The singular limit and its profile decomposition

This section carries the new content. The instance supplies a small parameter—in the entry problem the thin-atmosphere ratio $\varepsilon_{\text{atm}} = H_s/R_e \rightarrow 0$, where H_s is the atmospheric scale height and R_e the planetary radius—and the question is what the occupation-measure family does in that limit.

2.1 Loss of tightness and the defect measure

Rescaled by the phugoid-to-secular time ratio (which, as in the classical Chapman–Loh variables, enters as $\sqrt{\varepsilon_{\text{atm}}}$, not ε_{atm} [15, 16]), the occupation-measure family *is not tight*. Mass concentrates onto the low-dimensional equilibrium-glide slow manifold $\Lambda = 1$. In the 3-DOF (point-mass) approximation this is the scalar condition $L \cos \sigma / [m(g - V^2/r)] = 1$; in the full 6-DOF formulation the lift component $L \cos \sigma$ is not an independent input but is determined by the attitude quaternion through the body-to-LVLH rotation, so the slow manifold is a single scalar constraint on the 13-dimensional translational-rotational state. The full dynamics carry J_2 oblate gravity throughout: the gravitational acceleration has both a radial component $g_D(r, \phi)$ and a northward component $g_N(r, \phi)$, and the general slow-manifold condition retains both. The simplified scalar $\Lambda = 1$ is the leading-order reduction under spherical gravity and small flight-path angle. Retaining J_2 is not a refinement but a necessity, and for a reason sharper than the size of the terms. During skip maneuvers the northward component g_N produces a heading drift whose sign is fixed by the hemisphere, so it accumulates across successive skips; the radial component g_D varies the centrifugal balance at re-entry between $+0.16\%$ and -0.32% of the leading gravitational term, changing sign at 35.3° latitude, so it accumulates only while the ground track stays on one side of that parallel and partially cancels across it. The two effects therefore have different sign structures, no single worst-case latitude bounds both at once, and the enclosure may be neither tightened by assuming they reinforce nor loosened by assuming they cancel.

The loss of tightness is read on the *defect measure* of the rescaled family. The occupation measures live in the cone $\mathcal{M}_+([0, T] \times \mathcal{X} \times \mathcal{U})$ of nonnegative Radon measures, topologized weakly-* as the dual of $C_0([0, T] \times \mathcal{X} \times \mathcal{U})$. Writing the weak-* limit as $\mu_\varepsilon \rightharpoonup \mu_0 + \nu$, with μ_0 the limiting glide-manifold measure, the *defect measure* $\nu \geq 0$ is the nonnegative remainder—the mass that fails to concentrate on the slow manifold as $\varepsilon_{\text{atm}} \rightarrow 0$. On ν the situation is the measure-theoretic analogue of Lions’ concentration alternative—vanishing, dichotomy, or concentration [17]. The relevant scaling group is the phugoid time dilation $t \mapsto t/\sqrt{\varepsilon_{\text{atm}}}$ that localizes each atmospheric pass, so the bubbles are concentration at a shrinking timescale and it is the dilation half of the principle—the limit-case modification—that applies, not the translation (escape-to-infinity) case [18].

2.2 Skeleton, bubbles, residual

Rather than restore compactness, we extract the profile decomposition the dichotomy hands us:

- a finite-dimensional *skeleton* S —the concentration component, the glide manifold as a graph over (r, V) in the corridor subspace, for which a rigorous characterization with quantitative accuracy estimates already exists [19];
- one *boundary-layer bubble* per skip event, the analogue of a concentrating profile: a compact impulse set $\mathcal{I}_j$ of dimension $\leq m$, carried by an internal scaling group at each atmospheric pass, with orthogonality read off the scale-and-core separation condition that governs concentrating profiles generally [20]. The J_2 oblate-gravity terms are retained in the bubble dynamics: the northward component g_N introduces a heading drift during each skip, and the variation of g_D with latitude modifies the centrifugal balance at re-entry. The two carry different sign structures across a multi-skip ground track, so both have to be enclosed term by term rather than bounded at a single worst-case latitude;

Objective 3: Wrapping-effect suppression in boundary-layer bubbles

Boundary-layer bubbles represent discrete atmospheric skip events as compact impulse sets $\mathcal{I}_j$ carried by internal scaling groups. Because our framework fixes time as the independent variable, integrating interval bounds through highly dynamic skip events will induce severe “wrapping effects” (artificial inflation of bounding boxes). Must develop advanced intersection geometries (zonotopes, Taylor models, etc.) integrated into the profile decomposition to prove geometries maintain tight corridor bounds through multiple skip events without the reachable set exploding prematurely.

- a *compact residual* with a quantitative bound, $O(\sqrt{\varepsilon})$ in the corridor metric, in a norm chosen so that the loss of compactness actually disappears rather than merely being hidden [21].

The number of bubbles is bounded *a priori*, $K \leq K_{\max}(Q_{\max})$, by the integrated heat-load budget (a trajectory can only afford so many skips [22, 23]). That derivation carries a dependency which has to be discharged rather than assumed: the heat-load budget is integrated from a stagnation-point correlation whose chemical-equilibrium assumption is weakest at high altitude, which is precisely where a skip peaks. The wrapping effect to be suppressed here is not new: for linear systems it is already quantified, and the term is first order in the time step and exponential in the horizon times the matrix norm [24].

2.3 Almost-orthogonality of the scaling groups

The multi-parameter separation of the three scaling groups (atmospheric, attitude, elastic) is the Bahouri–Gérard almost-orthogonality argument, transported to defect measures: two profiles are almost-orthogonal when their scale parameters diverge, so that their rescaled supports become asymptotically disjoint and their cross terms vanish in the weak-* pairing on $\mathcal{M}_+([0, T] \times \mathcal{X} \times \mathcal{U})$. Establishing this non-interaction for all three groups simultaneously is the one genuinely new obligation, and it sits on the critical path.

Objective 4: Bahouri–Gérard almost-orthogonality argument

The framework relies on the Bahouri–Gérard almost-orthogonality argument to separate the multi-parameter scaling groups (atmospheric, attitude, and elastic). Under interval aerodynamics, these groups couple with an adversarial, bounded uncertainty. We must prove the non-interaction theorem for the defect measures while accounting for interval bounds, guaranteeing that the gap between the upper and lower aerodynamic bounds does not cause mass to leak between scaling groups during atmospheric transition, preserving the compact residual bound $O(\sqrt{\varepsilon})$.

The argument worth transporting is the constructive one: splitting a bounded sequence into an oscillatory and a singular part at each scale forces those scales to be orthogonal, and iterating extracts a pairwise orthogonal family, so orthogonality is produced by the construction rather than assumed by it [25].

2.4 The certified bound

The target statement is

$$\mathcal{R}_{\text{cert}}(w) \subseteq \underbrace{\left(\Phi_S(\Theta) \right)}_{\text{skeleton}} \oplus \underbrace{\left(\bigoplus_{j=1}^K \mathcal{I}_j \right)}_{\text{bubbles}} \oplus \underbrace{\mathbb{B}(C_{\mathcal{R}}\sqrt{\varepsilon})}_{\text{residual}}, \quad (2.1)$$

uniformly over the admissible family generated by the uncertainty set, not along a nominal trajectory. The exact residual corridor constant $C_{\mathcal{R}}$ and the maximum bubble count $K_{\max}$ are derived as explicit, closed-form algebraic expressions directly from the physical corridor bounds and aerodynamic interval widths, completely independent of numerical optimization logic. In the spherical-LVLH coordinates the corridor bounds are box constraints on (r, V) , so the interval widths that enter $C_{\mathcal{R}}$ and $K_{\max}$ have no wrapping from corridor intersection—the constants inherit the tightness of the coordinate choice. Requiring the constants in closed form, rather than as the output of a solver, is what separates a certificate from a computation: exact rational certificate representations are known to exist, with a documented bit-complexity that grows with the number of variables [26, 27].

Objective 5: Analytical derivation of corridor constants

The ultimate guarantee relies on deriving the residual corridor constant $C_{\mathcal{R}}$ and the maximum boundary-layer bubble count $K_{\max}(Q_{\max})$. The number of bubbles is bounded a priori by the integrated heat-load budget. We must derive explicit, closed-form algebraic expressions for $C_{\mathcal{R}}$ and $K_{\max}$ strictly from the physical corridor bounds (which are box constraints on (r, V) in the spherical-LVLH state), the initial conditions, and the aerodynamic interval widths, entirely without relying on numerical interior-point solvers.

3 The application: hypersonic entry

Entry is not an arbitrary illustration of the machinery above. It supplies the singular perturbation, the slow manifold and the concentrating events in one system, and it is a problem on which the exact methods are already known to fail.

3.1 Why the exact methods do not reach it

Exact computation means solving a Hamilton–Jacobi–Isaacs equation on a grid, which is exponential in state dimension and capped in practice near four or five states [28, 29]. A rigid-body entry vehicle carries thirteen states,

$$x = (r, \lambda, \phi, V, \gamma, \psi, q_0, q_1, q_2, q_3, p, q_b, r_b) \in \mathbb{R}^{13}, \quad (3.1)$$

and $17 + 2N_m$ once integrated heat load, wall temperature, ablation recession, mass loss and N_m aerothermoelastic modes are coupled [30]. The two counts use “degree of freedom” in its two customary senses, and it is worth separating them: the vehicle is six-degree-of-freedom in the configuration sense (three translational, three rotational), while Eq. (3.1) has thirteen coordinates spanning a twelve-dimensional state manifold once the unit-quaternion constraint $\|\vec{q}\| = 1$ is imposed.

So the exact grid-HJI method quits before the real problem starts. What makes the measure-theoretic route viable is not that it evades the dimension but that the bound it produces does not scale with it: the skeleton is low-dimensional by construction, the bubbles are compact, and the residual is a single ball.

3.2 The state space and the corridor

The translational state is carried in geocentric spherical position (r, λ, ϕ) and wind-frame velocity (V, γ, ψ) so that the entry-corridor constraints (bounded altitude, bounded speed, heating and load-factor limits) are box constraints on states, the natural setting for interval arithmetic. Attitude is a unit quaternion, free of Euler-angle singularities. Because interval arithmetic evaluates transcendentals natively, the traditional algebraic adjunctions required to force the system into a polynomial format [31] are obsolete. The state space carries only the physical dimensions—spherical position, wind-frame velocity, attitude quaternion, body rates, thermal, ablation, and modal states—with no artificial embeddings.

The shift to exact interval arithmetic renders traditional Savageau-style polynomial adjunctions—the density-square-root embedding $y = \sqrt{\rho/\rho_0}$ and the polynomial surrogate for wall temperature (Varma–Savageau–Voit recasting [31])—obsolete, as transcendentals are evaluated natively. These auxiliary states are eliminated entirely, collapsing the state space to its minimal physical dimensions and removing a source of wrapping. The coordinate system is settled: translational position in geocentric spherical (r, λ, ϕ) , planet-relative velocity in wind-frame components (V, γ, ψ) , and attitude as a unit quaternion. This choice makes the entry corridor a box in state space (tightening interval intersections) and keeps the equilibrium-glide slow manifold native. The trade-off—transcendental dynamics versus polynomial dynamics with nonlinear corridor intersection—is resolved in favor of the spherical formulation, where wrapping is distributed across time steps and controlled by step-size refinement rather than concentrated in corridor intersection.

4 Plan of work

Six concrete obligations carry the program; they are stated where each arises above and gathered here, in order, as the plan of work.

1. *Liouville equation for differential inclusions*: a measure embedding valid for the set-valued (interval) dynamics.

2. *Exact-arithmetic computational pipeline*: a deterministic interval integrator that verifies the bounds without floating point.
3. *Wrapping-effect suppression in boundary-layer bubbles*: tight enclosures carried through skip events without premature blow-up.
4. *Bahouri–Gérard under interval disturbances*: almost-orthogonality of the scaling groups under bounded uncertainty—the **critical path**, on which the profile decomposition and the closed-form bound rest.
5. *Analytical derivation of the corridor constants*: closed-form $C_{\mathcal{R}}$ and $K_{\max}$ from the physical corridor and interval widths, free of any numerical solver.
6. *State-space minimization*: elimination of the Savageau-style polynomial adjunctions now that transcendentals are evaluated natively.

Objectives 1–2 establish the interval occupation-measure embedding, 3–4 the profile decomposition, and 5–6 reduce it to closed form; objective 4 gates the rest.

4.1 Sequencing

The dependency structure suggests three phases rather than six parallel tracks. The first establishes the embedding—objective 1, with the density envelope of objective 2 that it needs in order to be stated on a concrete inclusion. The second attacks the critical path: objective 4, which is where the program either works or does not, and which should therefore be reached early enough that a negative result is still recoverable. Objective 3 runs alongside it, since the almost-orthogonality argument and the enclosure geometry constrain each other. The third phase reduces the decomposition to closed form—objectives 5 and 6—and completes the exact-arithmetic pipeline, which is the only deliverable that cannot be shortened by a weaker theorem.

If objective 4 resists a proof for all three scaling groups simultaneously, the fallback is to establish it for the atmospheric and attitude groups and to treat the elastic group as a widened interval rather than a separate profile. That weakens the residual constant without invalidating the decomposition, and it is worth identifying now as the retreat position.

References

- [1] J.-B. Lasserre et al. “Nonlinear Optimal Control via Occupation Measures and LMI Relaxations”. In: *SIAM Journal on Control and Optimization* 47.4 (2008), pp. 1643–1666.
- [2] R. B. Vinter. “Convex Duality and Nonlinear Optimal Control”. In: *SIAM Journal on Control and Optimization* 31.2 (1993), pp. 518–538.
- [3] D. Henrion and M. Korda. “Convex Computation of the Region of Attraction of Polynomial Control Systems”. In: *IEEE Transactions on Automatic Control* 59.2 (2014), pp. 297–312. DOI: [10.1109/TAC.2013.2283095](https://doi.org/10.1109/TAC.2013.2283095).
- [4] J. M. Ball. “A Version of the Fundamental Theorem for Young Measures”. In: *PDEs and Continuum Models of Phase Transitions*. Ed. by M. Rascle, D. Serre, and M. Slemrod. Vol. 344. Lecture Notes in Physics. Berlin: Springer, 1989, pp. 207–215.
- [5] A. F. Filippov. *Differential Equations with Discontinuous Righthand Sides*. Ed. by F. M. Arscott. Mathematics and Its Applications (Soviet Series). Dordrecht: Kluwer Academic Publishers, 1988. ISBN: 902772699X. DOI: [10.1007/978-94-015-7793-9](https://doi.org/10.1007/978-94-015-7793-9). Translated from the Russian.

- [6] J.-P. Aubin and A. Cellina. *Differential Inclusions: Set-Valued Maps and Viability Theory*. Vol. 264. Grundlehren der mathematischen Wissenschaften. Berlin: Springer-Verlag, 1984. DOI: [10.1007/978-3-642-69512-4](https://doi.org/10.1007/978-3-642-69512-4).
- [7] Z. Artstein. “Invariant Measures of Differential Inclusions Applied to Singular Perturbations”. In: *Journal of Differential Equations* 152.2 (1999), pp. 289–307. DOI: [10.1006/jdeq.1998.3527](https://doi.org/10.1006/jdeq.1998.3527).
- [8] Z. Artstein and V. Gaitsgory. “The Value Function of Singularly Perturbed Control Systems”. In: *Applied Mathematics and Optimization* 41.3 (2000), pp. 425–445. DOI: [10.1007/s002459910022](https://doi.org/10.1007/s002459910022).
- [9] S. Prajna, A. Jadbabaie, and G. J. Pappas. “A Framework for Worst-Case and Stochastic Safety Verification Using Barrier Certificates”. In: *IEEE Transactions on Automatic Control* 52.8 (2007), pp. 1415–1428. DOI: [10.1109/TAC.2007.902736](https://doi.org/10.1109/TAC.2007.902736).
- [10] P. Roux, Y.-L. Voronin, and S. Sankaranarayanan. “Validating Numerical Semidefinite Programming Solvers for Polynomial Invariants”. In: *Formal Methods in System Design* 53.2 (2018), pp. 286–312. DOI: [10.1007/s10703-017-0302-y](https://doi.org/10.1007/s10703-017-0302-y).
- [11] H. Peyrl and P. A. Parrilo. “Computing Sum of Squares Decompositions with Rational Coefficients”. In: *Theoretical Computer Science* 409.2 (2008), pp. 269–281. DOI: [10.1016/j.tcs.2008.09.025](https://doi.org/10.1016/j.tcs.2008.09.025).
- [12] R. E. Moore, R. B. Kearfott, and M. J. Cloud. *Introduction to Interval Analysis*. Vol. 110. Other Titles in Applied Mathematics. Philadelphia: Society for Industrial and Applied Mathematics, 2009. ISBN: 978-0-89871-669-6. DOI: [10.1137/1.9780898717716](https://doi.org/10.1137/1.9780898717716).
- [13] N. S. Nedialkov, K. R. Jackson, and G. F. Corliss. “Validated Solutions of Initial Value Problems for Ordinary Differential Equations”. In: *Applied Mathematics and Computation* 105.1 (1999), pp. 21–68. DOI: [10.1016/S0096-3003\(98\)10083-8](https://doi.org/10.1016/S0096-3003(98)10083-8).
- [14] L. Jaulin et al. *Applied Interval Analysis: With Examples in Parameter and State Estimation, Robust Control and Robotics*. London: Springer-Verlag, 2001. ISBN: 978-1-4471-1067-5. DOI: [10.1007/978-1-4471-0249-6](https://doi.org/10.1007/978-1-4471-0249-6).
- [15] D. R. Chapman. *An Approximate Analytical Method for Studying Entry into Planetary Atmospheres*. Tech. rep. Technical Note 4276. Ames Aeronautical Laboratory, Moffett Field, California: National Advisory Committee for Aeronautics, May 1958. 103 pp. Reissued as NASA TR R-11.
- [16] W. H. T. Loh. *Re-entry and Planetary Entry Physics and Technology. Dynamics, Physics, Radiation, Heat Transfer and Ablation*. Ed. by W. H. T. Loh. Vol. 1. Applied Physics and Engineering: An International Series. New York: Springer-Verlag, 1968. ISBN: 978-3-642-48929-7. DOI: [10.1007/978-3-642-48927-3](https://doi.org/10.1007/978-3-642-48927-3).
- [17] P.-L. Lions. “The Concentration-Compactness Principle in the Calculus of Variations. The Locally Compact Case, Part 1”. In: *Annales de l’Institut Henri Poincaré, Analyse Non Linéaire* 1.2 (1984), pp. 109–145.
- [18] P.-L. Lions. “The Concentration-Compactness Principle in the Calculus of Variations. The Limit Case, Part 1”. In: *Revista Matemática Iberoamericana* 1.1 (1985), pp. 145–201.
- [19] P. Lu. “Asymptotic Analysis of Quasi-Equilibrium Glide in Lifting Entry Flight”. In: *Journal of Guidance, Control, and Dynamics* 29.3 (2006), pp. 662–670. DOI: [10.2514/1.15384](https://doi.org/10.2514/1.15384).

- [20] P. Gérard. “Description du Défaut de Compacité de l’Injection de Sobolev”. In: *ESAIM: Control, Optimisation and Calculus of Variations* 3 (1998), pp. 213–233. DOI: [10.1051/cocv:1998107](https://doi.org/10.1051/cocv:1998107). In French.
- [21] S. Solimini. “A Note on Compactness-Type Properties with Respect to Lorentz Norms of Bounded Subsets of a Sobolev Space”. In: *Annales de l’Institut Henri Poincaré, Analyse Non Linéaire* 12.3 (1995), pp. 319–337.
- [22] Y.-Y. Shi, L. Pottsepp, and M. C. Eckstein. “A Matched Asymptotic Solution for Skipping Entry into Planetary Atmosphere”. In: *AIAA Journal* 9.4 (Apr. 1971), pp. 736–738.
- [23] A. J. Eggers Jr., H. J. Allen, and S. E. Neice. *A Comparative Analysis of the Performance of Long-Range Hypervelocity Vehicles*. Tech. rep. Report 1382. Ames Aeronautical Laboratory, Moffett Field, California: National Advisory Committee for Aeronautics, 1958.
- [24] C. Le Guernic and A. Girard. “Reachability Analysis of Linear Systems Using Support Functions”. In: *Nonlinear Analysis: Hybrid Systems* 4.2 (2010), pp. 250–262. DOI: [10.1016/j.nahs.2009.03.002](https://doi.org/10.1016/j.nahs.2009.03.002).
- [25] H. Bahouri and P. Gérard. “High Frequency Approximation of Solutions to Critical Nonlinear Wave Equations”. In: *American Journal of Mathematics* 121.1 (1999), pp. 131–175. DOI: [10.1353/ajm.1999.0001](https://doi.org/10.1353/ajm.1999.0001).
- [26] E. L. Kaltofen et al. “Exact Certification in Global Polynomial Optimization via Sums-of-Squares of Rational Functions with Rational Coefficients”. In: *Journal of Symbolic Computation* 47.1 (2012), pp. 1–15. DOI: [10.1016/j.jsc.2011.08.002](https://doi.org/10.1016/j.jsc.2011.08.002).
- [27] V. Magron and M. Safey El Din. “On Exact Reznick, Hilbert-Artin and Putinar’s Representations”. In: *Journal of Symbolic Computation* 107 (2021), pp. 221–250. DOI: [10.1016/j.jsc.2021.03.003](https://doi.org/10.1016/j.jsc.2021.03.003).
- [28] S. Bansal et al. “Hamilton-Jacobi Reachability: A Brief Overview and Recent Advances”. In: *56th IEEE Conference on Decision and Control (CDC)*. 2017, pp. 2242–2253. DOI: [10.1109/CDC.2017.8263977](https://doi.org/10.1109/CDC.2017.8263977).
- [29] I. M. Mitchell, A. M. Bayen, and C. J. Tomlin. “A Time-Dependent Hamilton-Jacobi Formulation of Reachable Sets for Continuous Dynamic Games”. In: *IEEE Transactions on Automatic Control* 50.7 (2005), pp. 947–957. DOI: [10.1109/TAC.2005.851439](https://doi.org/10.1109/TAC.2005.851439).
- [30] J. J. McNamara and P. P. Friedmann. “Aeroelastic and Aerothermoelastic Analysis in Hypersonic Flow: Past, Present, and Future”. In: *AIAA Journal* 49.6 (2011), pp. 1089–1122. DOI: [10.2514/1.J050882](https://doi.org/10.2514/1.J050882).
- [31] M. A. Savageau and E. O. Voit. “Recasting Nonlinear Differential Equations as S-Systems: A Canonical Nonlinear Form”. In: *Mathematical Biosciences* 87.1 (1987), pp. 83–115. DOI: [10.1016/0025-5564\(87\)90035-6](https://doi.org/10.1016/0025-5564(87)90035-6).